

PLASMA SPRAY PROCESSORS

Works Instruction

QF/QMS/02/40

Rev. 00

Date : 21/04/2025



Operations/Work/Job Activity covered by this assessment: **Raw material inspection-Thermal spray powder**

Required Personal Protective Equipment (PPE) :



Safety glasses must be worn at times.



Long and loose hair must be contained.



Appropriate footwear with substantial uppers must be worn.



Close fitting/protective clothing to cover arms and legs must be worn.



Respiratory protection devices required.



Oil free leather gloves and spats must be worn.



Hearing protection must be used when using this machine.

To ensure proper inspection, handling, and verification of thermal coating powder before approval for use.

Departments Involved: Stores / Quality Assurance (QA) / Lab

1. Material Receiving and Initial Check (Store Person) receive the thermal coating powder from supplier.
2. Check packaging and compare with packing list for damage or leakage, supplier name, quantity, powder name, date, batch/lot number.
3. Hold material in the **"Under Inspection"** area.
4. Inform the QA department for further inspection.
5. QA personnel log the material in the Raw Material Inspection Register, including: Date of receipt, Powder name and batch number, Quantity and check with material test certificate.
6. QA collects 1 box of powder from the received batch for validation.
7. Hand over the sample & powder to the spraying operator with valid parameter and ensure to perform the same.
8. Perform deposit efficiency and sample spraying on test coupons as per standard procedure.
9. Ensure correct parameters are used (e.g., spray distance, temperature, feed rate, Argon/Hydrogen flow, Nitrogen, Amp, Volt, Fuel, combustion pressure).
10. Conduct metallographic analysis on sprayed samples. Coating hardness, Porosity level, Microstructure examination.
11. Review test results and compare them with the specification/standard.
12. After completing the analysis, the records have to be properly maintained and documented.

Assessor	Prasanna Wadkar	Signature:		Date:	
Approved:	Haarsh Doshi	Signature:		Date:	